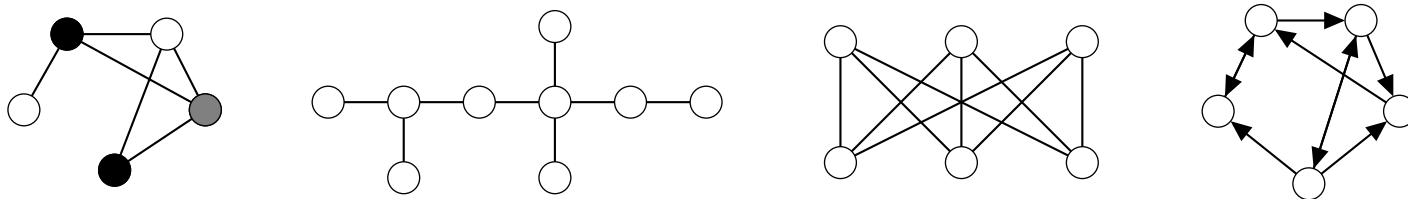


MAT 350: Graph Theory, Augsburg University, Spring 2026 A

All course policies subject to change with notice.



Where and when does class meet?

Class meets in person **MWF 9:20-10:30 AM in Hagfors 151**. Our first day of class is Wednesday January 21 and our last class is Wednesday May 6 during finals week. Classes might occasionally switch to Zoom with notice.

How do I communicate with the professor?

My name is Dr. Suzanne Dorée. You are welcome to call me Professor Dorée, Professor Su, or just Professor. The best way to reach me outside of class is to email me: doree@augsborg.edu using your Augsburg email account.

What will we study in this course?

This course explores concepts of graph theory including standard graphs, classic questions, and topics such as degree sequences, isomorphism, connectivity, traversability, matching, planarity, coloring, and graph metrics with a focus on theoretical structures. We will encounter the following breadth of topics, in varying depths.

- Graph models, applications, games, famous problems, and open problems.
- Basic graph vocabulary including bipartite graphs, adjacency matrices/lists, and complements.
- Standard graphs, famous graphs, subgraphs, and constructing new graphs.
- Degree sequences and regular graphs.
- Connectivity, paths and cycles, searches, components, cut vertices, and bridges.
- Distance, center of a graph, and dominating sets.
- Graph isomorphism and invariants, classification of graphs, automorphism.
- Trees, minimum spanning trees, encoding information in trees.
- Traversing graphs, Eulerian, and Hamiltonian.
- Matching and factoring graphs.
- Directed graphs, tournaments, and strong connectivity.
- Planarity, crossing number, and genus.
- Vertex coloring.

What will we learn in the course?

This course is designed to help you:

- Extend your knowledge of elementary graph theory.
- Deepen your understanding of equivalence, isomorphism, and alternate representation.
- Extend your mathematical reasoning skills: questioning, conjecturing, developing informal and formal proof, understanding algorithms, and geometric reasoning.

- Develop your mathematical reading skills and increase your comfort with the overall theoretical structure of mathematics: the role of definitions, examples, and theorems; abstracting concepts; and logical reasoning within a system.
- Strengthen your mathematical writing and speaking skills: appropriate use of mathematical language, notation, and symbolism; conventions and style in mathematical writing and speaking; and drawing detailed graphs by hand.
- Develop your computational reasoning skills and increase your comfort working with a programming language (Python) as an exploratory tool using simple code.

What academic requirements does this course count for?

This course counts as four upper division credit hours. With a grade of C- or higher (or Pass), this course counts as follows.

- Mathematics majors: option for theoretical structures requirement or advanced elective (MAT 350+), not both.
- Mathematics minor: option for advanced elective.
- Data science major: option for advanced elective (in pair with MAT 302 Discrete Mathematical Structures).
- Computer science majors (Fall 2024 catalog and later): option for advanced elective. If you are under an earlier catalog, ask your faculty mention which requirement this course will meet for you.

What background knowledge do I need to be ready for this class?

The prerequisite for this course is MAT 302 *Discrete Mathematical Structures* with a grade of C- or higher (or Pass). We will use discrete mathematics every day. If you received C+ or lower in the discrete class or if you are worried about taking a math class, please meet with me early in the semester and I will help you plan for success. No prior coding experience is expected, but it is useful.

What texts, materials, or technology do I need for this class?

We have two textbooks. Both are required and used nearly every class. The textbooks are available for purchase at the bookstore (or online), or you can access the library's e-Book free using links on our Moodle page.

FW "The fascinating world of graph theory," by Benjamin, Chartrand, and Zhang (Princeton, 2015)

GT "A first course in graph theory," by Chartrand and Zhang (Dover, 2012)

You will need regular access to the Internet to read email and to access resources on Moodle. If you do not have reliable Internet at home, contact the Info/Tech Desk at Lindell Library for help.

You will also need regular access to a device that scans written work; any cellphone with a camera has this functionality. The on-campus printers also scan, but not as well. Alternatively, you can use a writable electronic surface (such as a tablet).

How can I get help with this class?

There are two approved ways to get help with this class.

From me: I very much enjoy working with students outside of class and look forward to helping you learn. One way I do this is to hold office hours

MW 2:30-3:30 PM and F 2:30-3:00 PM

I am often, but not always, available during community time **MWF 12:30-1:00 PM**. You can stop by in person during office hours and we can work together on learning course material. You are also welcome to ask me about tips for success in the course or to chat about mathematics or college in general. During my office hours I am either in my office (Hagfors 185) or in the study area across the hall. No appointment is necessary. I know my office hours do not fit everyone's schedule so I am often available at other times – just ask! If I am not available when you need help, you are also welcome to e-mail me. I am pretty good at helping via email.

From classmates: You are encouraged to meet regularly with classmates, which means students currently enrolled in the course. Students often hang out and work on homework in the study area across from my office or in the Computer Science home up on third floor of Hagfors most of the day MWF.

How can I learn best in this class (or what should I do to get an A)?

Before class, print the class notes or open them in an annotation software on a device that can lay flat on a table (e.g. tablet). You can print free on campus, and if you run out of prints ask for more free prints. I want you to spend class time thinking about the mathematics, not frantically copying the boards. You should also print (or open) the corresponding assignment.

Being in class to learn with me and classmates is very important. We will work on learning the new materials by completing the class notes and working on activities. I will introduce new ideas, work through examples, help you do the activities, discuss important course policies, and remind you of important deadlines. You will not want to miss class! I understand that at times you might occasionally have a serious situation that leads to your missing class. Please try to avoid missing class in less serious situations. We will use all of class time so please arrive on time and stay for the full session.

Building a supportive classroom where you feel welcome is a priority for me and I ask your help making sure all students are welcome and included. I have various course policies to support this work. In particular, during lecture time I do not allow voluntary calling out of answers, asking questions, or unsolicited comments. When you have questions during lecture (which will happen), you can quietly ask your neighbor or write down your questions for office hours. I will randomly assign seats each day so that you have an opportunity to work with many different classmates. I encourage you to embrace these practices, even if they seem a bit new to you, as I believe they will help you succeed.

You earn points in this class in three categories: engagement, homework assignments, and quizzes. The primary way to get engagement points is to attend class. At times I offer a chance to earn extra credit engagement points by doing things that I believe can help you learn such as attending office hours or working with a study group. It would be wise to take advantage of these extra credit offers!

The majority of assignment points come from homework assignments, which are posted on Moodle. These assignments give you a chance to get feedback on how well you are learning the material. To start you should print out the assignment (or you can open the PDF of the assignment on an electronic device in annotation software). In either case, your solutions should be handwritten. Using these standardized pages keeps your work organized and allows you to skip a problem and come back as needed.

The homework assignments are long and cover one or several days of material, so it will often take several study sessions to complete an assignment. Some of the problems measure your understanding of the lecture and reading, others ask you to explain key ideas or practice key skills. Other problems show you applications, give you practice reading and writing proofs, or introduce ideas beyond lecture. While many of the problems are relatively straightforward, I like to include more challenging problems. Do not be surprised if you get stuck or are not able to complete every problem fully without help from classmates or me. That is okay. You are welcome to submit incomplete assignments so you get feedback on whatever you could do.

You are welcome to work with classmates on assignments or to ask me questions during office hours or via email. If you work with a classmate, write up the final solution yourself to make sure you understand it. Please do not copy from classmates, look for solutions online, or otherwise compromise the integrity of your work. Plus, those strategies might get an assignment “done” but they will not help you learn the material.

Submit your work on homework assignments on Moodle. If you are writing on paper, scan (not photo) the pages into a single PDF, edit the image as needed to be readable, and upload to Moodle. If you are annotating the PDF, export to PDF (to capture your work), proofread your document, and upload to Moodle.

Homework assignments are typically due Sundays and Tuesdays by 11:59 PM, although they are accepted through 4:59 PM the following day without penalty. At 5:00 PM that day the solutions will appear on Moodle. Be sure to submit whatever work you have well in advance of when the solutions appear because if you do not submit anything, the maximum you can get on the assignment is 50%.

You will then have a few days to grade your original work, then revise and correct any problems based on the solutions and grade your revised work. The revised/graded version of the assignments are typically due Thursdays by 11:59 PM, although they are accepted through the start of class on the day of the corresponding quiz without penalty. Grading your own assignments is the best way to reflect on your own work and learn from any mistakes you might have made. Try to have fun with it. You are welcome to work with classmates on the grading or ask me questions about it.

There are three practices that will save you time and increase how well you learn the material.

- Start the homework assignment right away after class, at least to fill in whatever you can from what we did in class. The assignment might not be due for awhile, but get into this practice.
- When you return to work the homework assignment, start by reading the textbooks to understand the material better. There are two textbooks: FW is relatively easy to read, GT is more comprehensive and technical. Often it is helpful to read the FW portion first to get the big ideas and then GT to get the details.
- Try to work on graph theory every day. Waiting a few days makes it easy to forget what you've learned.

A few times during the semester I will offer a chance to earn extra credit assignment points by doing things that extend your understanding of course material (beyond what's on quizzes). It would be wise to take advantage of these extra credit offers too!

A couple of times during the semester, I will give you a speaking assignment that asks you to prepare and give a short presentation to class. There is a serious penalty for not giving assigned presentations. If you must miss class on a presentation day, be sure to ask me about make-up options.

There is one additional type of assignment – assignment infinity. It is important to me that you engage in independent work that extends beyond course material at some point during the term. My expectation is that you complete **20 points** worth of independent work – 10 points by March 6 and 10 points by April 17. There is also a serious penalty for not submitting anything for assignment infinity, so make sure you do something.

The last measure of your learning is weekly quizzes, typically on Fridays. These quizzes cover the topics from class and the assignments from about one week's worth of material. Quizzes are closed book and individual – time to show me what you have learned. No reference material, computational technology, or collaboration is allowed on the quizzes, unless you have special needs that we have made prior arrangements to accommodate through our CLASS program. There are no “exams” in this class. Think of the quizzes as “mini-exams” and actually study for them. The last quizzes are during our assigned final exam time.

How is my work graded?

If you are in class from start to finish and paying attention, you get five engagement points for the day. If you arrive a minute or two late, step out during class, or leave class a minute or two early you usually get four points. If you miss more than a couple of minutes of class, you get two points. If you miss class entirely, you get zero points. If you have an emergency or absolutely unavoidable situation that causes you to miss more than a few classes during the term, be sure to ask me if there is any way to make up some of the missed points.

You will be grading your own homework assignments.

- The first step in grading each problem (meaning all of the parts of one problem) is to read the posted solutions and other allowed resources – which means: class notes, the textbooks, anything posted to our Moodle page, classmates, and me. You are also allowed to use AI to ask for a solution or additional explanation, but be aware that AI's solution may be incomplete or use concepts or techniques beyond what we have learned in the class.
- Next, evaluate your work on that problem using the scale:
 - 5 Correct, complete, and clearly written solution.
 - 4 Nearly correct, complete, and clearly written solution.
 - 3 Some correct, relevant progress toward a solution OR correct answer with no work.
 - 2 Work shows understanding of the problem but little or no progress toward a solution.
 - 1 Attempted but does not show understanding of the problem.
 - 0 Not attempted.
- For scores of 0, 1, 2, or 5 no comments are required. For scores of 3 or 4, include some brief comment about what is incorrect, missing, or unclear.
- Next, use the posted solutions and other allowed resources to correct and revise your solution. Please write your revisions in a red/pink ink (and do not use those colors for your original work). If you already had a 5, then there is nothing to do. If you had a 3 or 4, please keep work that was correct and add/fix the rest. If you had a 0, 1, or 2 you should redo the problem and it is okay to copy my solutions, hopefully with them making sense to you.

- Then, grade your revised work using the same scale. Typically you will have a 5 for this second grade. Note that if you don't want to correct your original work (or run out of time), then you will may have the same number again.
- At this point in the process each problem should have score written to the left of the work in the format **3+5**. Please keep the score in that format (versus writing 8). Note that if you did not submit the work by the original due date that your first score is a 0, but **0+5** is still possible. I strongly encourage you to submit whatever you have completed in time for it to count towards that first score.
- Last, calculate your grade on the entire assignment using the following formula, recalling the ceiling function rounds up to the nearest integer:

$$\left\lceil \frac{\text{Total Points Earned}}{\text{Number of Problems}} \right\rceil$$

Please write the entire formula and grade on the top of your page. For example, if you earned a total of 99 points on 12 problems, you would write $\left\lceil \frac{99}{12} \right\rceil = 9$. Note that the max score possible is 10.

If you submit the original assignment but do not grade it, I will assign a grade of at most 5 points without feedback on individual problems. I reserve the right to limit how often I mark ungraded work. At times I will review your assignments and grading and reserve the right to change the grade if there is a major discrepancy. You are welcome to ask me to review an assignment during office hours with you.

I grade the speaking assignments out of 10 points as follows:

- 10** Organized, prepared, and practiced presentation with clear explanations and visual aids on assigned topic. Mathematics is completely correct or there was one small error or omission.
- 8** More than one small error/omission or one larger error/omission.
- 6** Reasonable effort.
- 20** Did not give presentation.

I grade assignment infinity in units of 10 points (all or nothing) but if you do work early enough you may revise it if needed. You are each expected to complete 20 points on assignment infinity. If you do more, it counts as extra credit. If you do not do any, you are penalized by getting a 0 on the assignment and another -50 points. I'm not kidding!

Each quiz is graded out of 10 points per unit. Most quizzes cover two units and are, therefore, worth 20 points. Our final quiz covers three units and is worth 30 points.

How is my course grade calculated?

First, I calculate the percentage of points you earned in each category and compute the weighted average with

- 30% Engagement
- 30% Homework Assignments
- 40% Quizzes

Next, I convert your percentage grade to a letter grade using the following specifications.

- A requires at least 94% overall and at least 84% on each category.
- A- requires at least 91% overall and at least 81% on each category.
- B+ requires at least 87% overall and at least 77% on each category.
- B requires at least 80% overall and at least 70% on each category.
- B- requires at least 77% overall and at least 67% on each category.
- C+ requires at least 73% overall and at least 63% on each category.
- C requires at least 66% overall and at least 56% on each category.

- C- or P requires at least 63% overall and at least 53% on each category. This grade is the minimum required to count the course towards the mathematics major or minor, computer science major or minor, data science major, for Education programs, or as a prerequisite for any CSC, DST, or MAT course.
- D or LP requires at least 52% overall and at least 42% on each category. This grade is the minimum required to receive credit towards graduation for the course.
- Any other situation is graded F or N.

You are welcome to take this course P/LP/N but you should check with your faculty mentor (or your degree audit) first because there is a limit to the number of P/LP/N courses overall, towards a major, and towards a minor. Some students like this option because the grades of P, LP, and N do not count into your Grade Point Average (GPA).

I reserve the right to lower your course grade if your attendance, punctuality, participation, preparation, or behavior falls well below the expectations listed in the next section. If during the term you are not progressing towards a passing grade or if your attendance, participation, preparation, or behavior is falling well below expectations, I will usually email you or send an Academic Progress Report.

I reserve the right to lower your course grade if you violate the Academic Honesty Policy in this course. If that happens, I will notify you in writing. If you are ever in a situation where you are not sure if something is allowed, please ask!

What is expected of me?

I expect you to:

- Read your Augsburg emails regularly, including reading emails I send directly or via Moodle, and access our Moodle site as needed.
- Attend and participate in every class. Be on time and prepared for every class. Arrange to stay for the entire class period. If you must come late or leave early on special occasion, please let me know in advance. Avoid scheduling appointments that overlap class time. If you have an occasional excused absence, send me an email as soon as possible so that I know you are okay. You do not need to tell me why you are late or absent, just check in.
- Pay attention during class. The only thing you should be reading, viewing, writing, or discussing during class is the current material. Use of technology is important for learning in this class, but it can also be a huge distraction. To help you focus, you may not be using your cell phone, laptop, tablet, or other technology during class unless directed by me to do so. At such times, I will let you know that you may use technology for a task. You are only allowed to use your device to work on the assigned task. (For example, no watching videos or completing other assignments.) One exception is that you are allowed to use a tablet or laptop laid flat on the table during class at any time for the purpose of taking course notes, but not for anything other than taking course notes. Another exception is that you are allowed to use a cell phone to photograph my notes or your own work at the whiteboards. If you do not comply with this policy, I reserve the right to ask you to leave class. Exceptions to this policy require advanced approval by the instructor, for example, for a disability-related accommodation or in case of a family emergency.
- Do all of the assigned work. Plan on spending 9-12 hours a week outside of class time working on the course. Try scheduling at least three hours of work time (perhaps one hour + two hours) between every pair of classes.
- Submit work when due. All work should be uploaded to Moodle by the indicated date and time. You may submit work for as long as the Moodle link allows. Even if you miss class, work is still due then. I reserve the right to limit how many and which assignments are eligible for late submission.
- Get help when you need it. I know it is hard to ask for help sometimes, but I really enjoy helping students to learn so please reach out to me for help as you need it. You would be wise to arrange to meet with a classmate outside of class to compare work as well. Do not wait until the last minute to get help or meet with classmates. If you did not do well in previous college mathematics courses, or it has been a few years since you took those courses, you should make every effort to arrange your schedule to be able to attend office hours.
- Take quizzes and give presentations when scheduled, including during finals week. If you have a potential unavoidable conflict with a date, discuss the options with me well in advance and I will try to arrange a make-up. If you miss a quiz or presentation without having made prior arrangements, get in touch with me as soon as possible to ask if you may take a make-up quiz. While I often allow make-up quizzes, understand that a missed quiz may result in 0 points. I reserve the right to limit the number of make-up quizzes and to designate the time period during which make-up quizzes are offered. I also reserve the right to require follow-up conversation with me during office hours to check on your understanding of the material for any quiz.

- Work with others in the class when asked to do so. Work alone when asked to do so. Follow all academic honesty rules. To be clear, you are encouraged to share ideas, check answers, and compare solutions or explanations with classmates, but any work you submit must be "your own work" as defined by the Augsburg College Academic Honesty Policy. In particular, your original work on homework assignments must be your own work. For a copy of Augsburg University Academic's Honesty Policy, see <https://tinyurl.com/AcademicHonestyAugsburg> The consequence of a violation of this policy is anywhere from a 0% on the work to an automatic F in the course. I am also required to notify the Dean of the School of Natural Sciences who may impose further consequences. Please protect your reputation and uphold your integrity by committing to being academically honest throughout this course.
- Check your grades on Moodle gradebook (within our course page) regularly and email me if you have any questions about your grades. Be sure to check for missing assignments by looking at the full list of grades, not just the course percentage.
- Keep me up-to-date on how the class is going for you. Be sure to come in and talk with me if I email you or send you a progress report so that we can get you back on track.
- Let me know of any special needs or concerns you have. If you need accommodations because of a disability, please contact the CLASS Office – see below.
- Creating a classroom environment that welcomes and values all students is very important to me. I ask your assistance in helping support one another by following classroom rules. If you feel unwelcome or unvalued at any time, or have any suggestions for improvement, please let me know so I can try to help. Please do not be shy about telling me what is or is not working for you.
- Stay home if you are sick with a contagious virus or aren't sure why you're feeling sick. But please drop me an email to let me know.
- As with all Augsburg classes, you should understand and comply with all University policies described in the University catalog and the Student Guide: <http://inside.augsburg.edu/studentaffairs/studentguide/>.
- Complete any in-class and online course evaluation forms.

Where can I get help or accommodations?

Your success in this class is important to me. Here are some university resources that can help.

- **Disability Accommodations and Accessibility:** If you need disability-related accommodations to have equitable access in this course, please contact the CLASS Office (Disability Resources) at class@augsborg.edu or schedule a meeting with CLASS at www.augsburg.edu/class. Phone: (612) 330-1053. Location: Lindell Library, 216. If accommodations are required, the CLASS Office will notify me privately about your needs. Please note that you will not be required to disclose your disability to me, only your accommodations needs. The sooner you let me know your needs, the sooner I can assist you in achieving your learning goals in this course.
- **Health and Wellness:** As part of the University's support systems, you have access to campus resources to maintain your safety, health, and well-being. We understand that as a student you may experience a range of issues that can cause barriers to learning. These stressful moments can impact academic performance or reduce your ability to engage. If you or someone you know are suffering from any challenges, you should reach out for support. For concerns about physical or mental wellbeing, the Center for Wellness and Counseling is located on the first floor of Anderson Residence Hall, 612-330-1707 or <https://www.augsburg.edu/cwc/>.
- **Augsburg's Student Support Services** maintains a Student Support Guide that lists community resources for crisis support, academic support, personal concerns, health concerns, basic living needs, immigration and legal support. Check out the Auggie Basics web page for more info.
- **Title IX and Sex Discrimination:** Augsburg University is committed to creating a safe learning environment for all students, including one that is free of sex misconduct. If you or someone you know has experienced sexual misconduct, including sexual assault, dating violence, domestic violence, and stalking, you may contact Mike Grewe, Dean of Students and our Title IX Coordinator, at titleix@augsborg.edu or 612-330-1499 to report an incident, seek support, and/or take action. Please be aware that faculty members are required to disclose information about suspected or alleged sexual misconduct or other potential violations of the Augsburg University Sexual Misconduct Policy to the Title IX Coordinator. If the Title IX Coordinator receives information about an incident, they will reach out to offer information about resources, rights, and procedural options as a member of the campus community. If you or another student you know wishes to speak to a confidential resource who does not have this reporting responsibility, you may contact the Center for Wellness and Counseling at cwc@augsborg.edu or 612-330-1707; Campus Ministry at 612-330-1732; the Aurora Center at aurora@umn.edu or 612-626-9111 (24/7); or the 24/7/365 Crisis Support Line at 844-287-6963.
- **Supporting Pregnant and Parenting Students:** Augsburg students who are pregnant or are experiencing related conditions (including childbirth, termination of pregnancy, or lactation; related medical conditions; or recovery therefrom) may request reasonable modifications. Students may contact the Title IX Coordinator at titleix@augsborg.edu or 612-330-1034 to learn about specific actions the University can take to prevent discrimination and ensure access to Augsburg's educational programs and activities. Augsburg University also has a designated student parent navigator for parenting students needing assistance in meeting their family's basic needs, Noah Greenfield, Program Coordinator in the Dean of Students Office, at deanofstudents@augsborg.edu or 612-330-1160.

What are the university's policies on privacy?

As your instructor, I am committed to protecting your privacy by adhering to the Family Educational Rights and Privacy Act (FERPA). This includes using your educational data only as allowed by FERPA. Similarly, you should not share classmates' personal information (such as their name, phone number, email, address, pictures, etc.) without their explicit consent. Do not photograph, duplicate, digitally or verbally share representations of other students' written or creative work, classroom dialogue, discussion posts, or presentations on social media or with any other third party. If you have questions about these policies please ask me for additional clarification.

Schedule – subject to change with notice

Monday	Wednesday	Friday
MAT 350: Graph Theory Spring 2026	January 21 1: Modeling with graphs	January 23 2: Order, size, and degrees
January 26 3: Standard graphs and representing graphs (Assn 1 due)	January 28 4: Complements, unions, and joins (Assn 2 due)	January 30 5: More products Mini quiz 1 + Explain Assn ∞
February 2 Applications of graph theory talks (Assn 3 due)	February 4 Introduction to Computing with Graphs (Assn 4 due)	February 6 Introduction to My Graph Family Quiz 2-3
February 9 6: Regular graphs (Assn 5 due)	February 11 7: Degree sequences (XC 5.5 extra due)	February 13 8: Subgraphs, part 1 Quiz 4-5 + Assn ∞ 1 topic
February 16 8: Subgraphs, part 2 (Assn 6 due)	February 18 9: Isomorphism, part 1 (Assn 7 due)	February 20 9: Isomorphism, part 2 Quiz 6-7 + Assn ∞ 1 schedule meeting
February 23 10: Automorphism (Assn 8 due)	February 25 11: Walks, paths, circuits, and cycles (Assn 9 due)	February 27 12: DFS, BFS, distance, and geodesics, part 1 + Mini quiz 8
March 2 12: DFS, BFS, distance, and geodesics, part 2 (Assn 10 due)	March 4 13: Eccentricity, radius, and diameter (Assn 11 due)	March 6 14: Center and more about distance Quiz 9-10 + Assn ∞ 1 due
March 9 15: Domination and location numbers, part 1 (Assn 12 due)	March 11 15: Domination and location numbers, part 2 (Assn 13 due)	March 13 16: Connected graphs, components, and bridges, part 1 Quiz 11-12 + Assn ∞ 2 topic
March 16-20 No classes: Spring break		
March 23 16: Connected graphs, components, and bridges, part 2 (Assn 14 due)	March 25 17: Trees (Assn 15 due)	March 27 18: Connected graphs, degrees, and cut vertices, part 1 Quiz 13-14 + Assn ∞ 2 schedule meeting
March 30 Trees talks (Assn 16 due)	April 1 18: Connected graphs, degrees, and cut vertices, part 2 Quiz 15-16	April 3 No classes: Good Friday
April 6 19: Eulerian and Hamiltonian graphs, part 1 (Assn 17 due, XC 17.5 due)	April 8 19: Eulerian and Hamiltonian graphs, part 2 (Assn 18 due)	April 10 20: Directed graphs and orientation Quiz 17-18 + Assn ∞ 2 due
April 13 21: Bipartite graphs (Assn 19 due)	April 15 22: Matching, factors, and the Petersen graph, part 1 (Assn 20 due)	April 17 22: Matching, factors, and the Petersen graph, part 2 Quiz 19-20 + Assn ∞ 3 topic, if approved
April 20 23: Planarity, part 1 (Assn 21 due)	April 22 23: Planarity, part 2 (Assn 22 due)	April 24 24: Crossing number and genus, part 1 Quiz 21-22 + Assn ∞ 3 schedule meeting
April 27 24: Crossing number and genus, part 2 (Assn 23 due)	April 29 25: Vertex coloring	May 1 Last day of class (Assn 24 due)
May 4 No classes: finals week (Assn 25 due)	May 6 Mega quiz 23-24-25 + Assn ∞ 3 due (optional)	Subject to change with notice Effective Feb 9